



Basics of Cyber-Security Networking

Networking principles: OSI model, MAC and IP addressing


```
mirror_mod = modifier_ob.  
#set mirror object to mirror  
mirror_mod.mirror_object =  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
  
#selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
  
print("please select exactly  
  
-- OPERATOR CLASSES --  
  
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"  
  
context):  
context.active_object is not
```

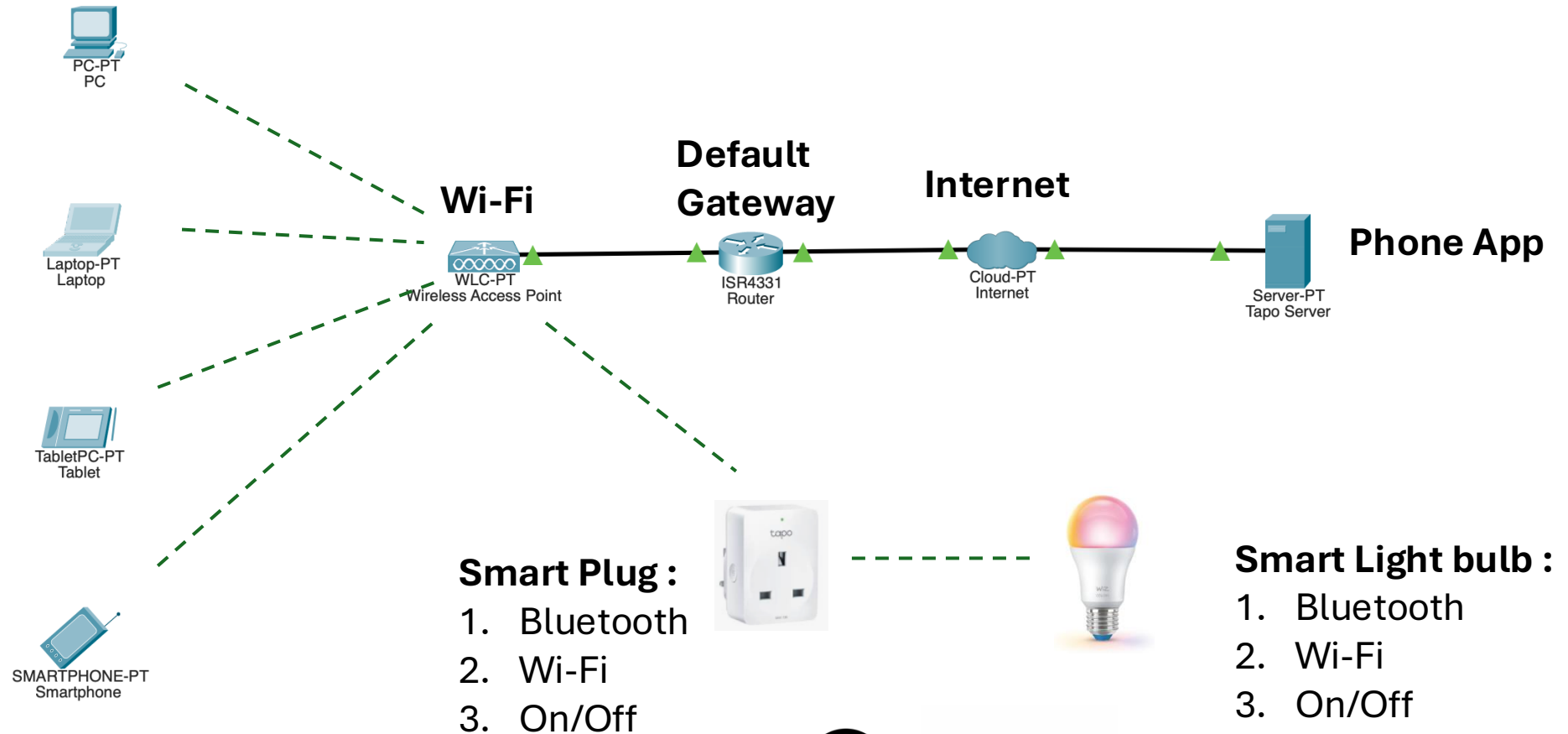
Ethical Hacking Statement

- A hacker is a person who **breaks into a computer system**.
- The reasons for hacking can be many:
 - installing malware, stealing or destroying data, disrupting service, and more.
- Hacking can also be done for **ethical reasons**, such as trying to find software vulnerabilities so they can be fixed.
- Ethical hacking involves the **legal use of hacking techniques** for benevolent versus malicious purposes.
- Ethical hackers use **penetration testing** and other tactics to find software vulnerabilities and other security weaknesses so they can be promptly addressed.
 - **Permission is required**
- <https://www.cisco.com/c/en/us/products/security/what-is-a-hacker.html>

The Network

Connectivity:

1. Bluetooth
2. Wi-fi
3. Internet



Hacker

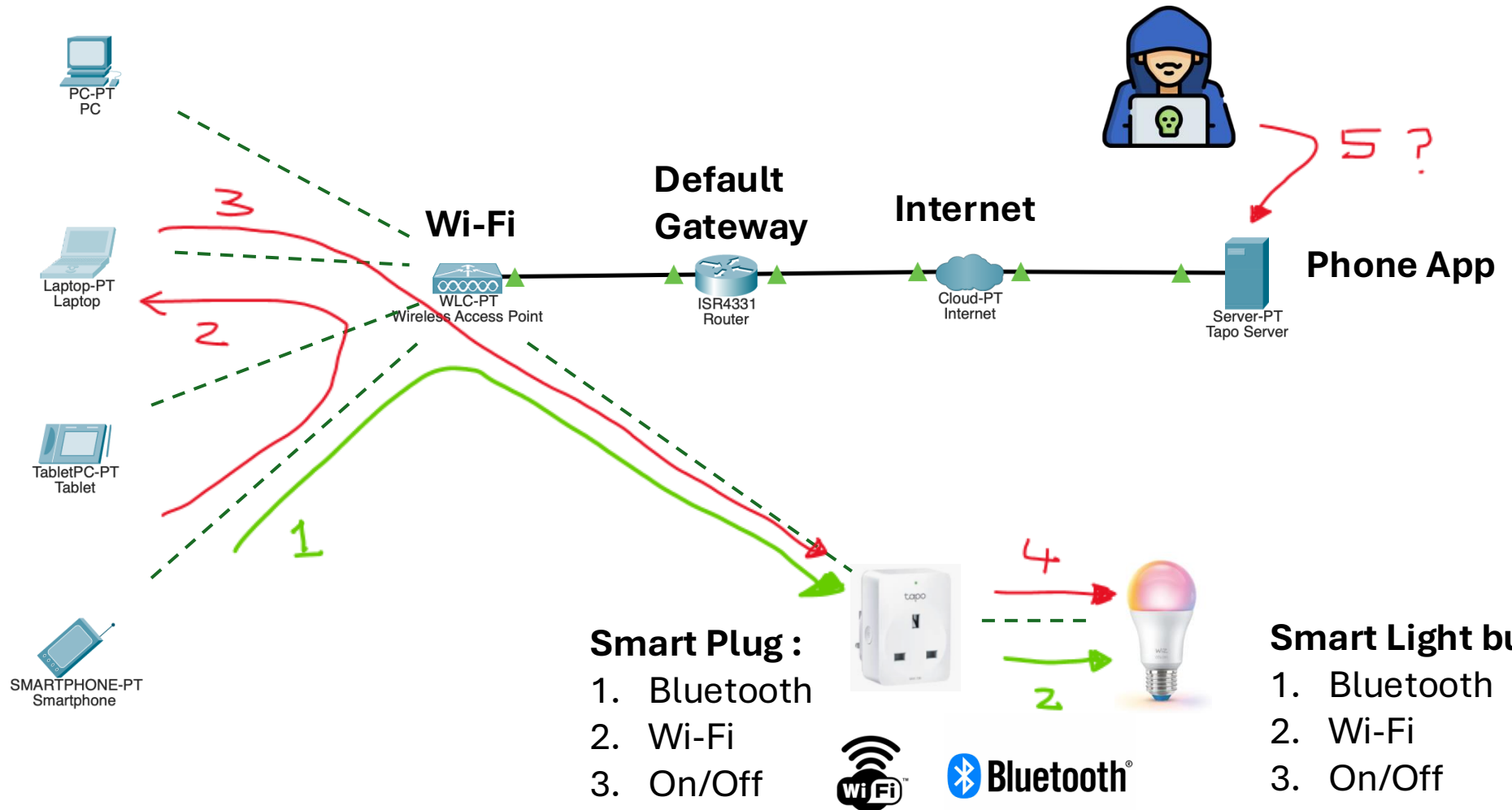


Reconnaissance:

- devices
- addressing

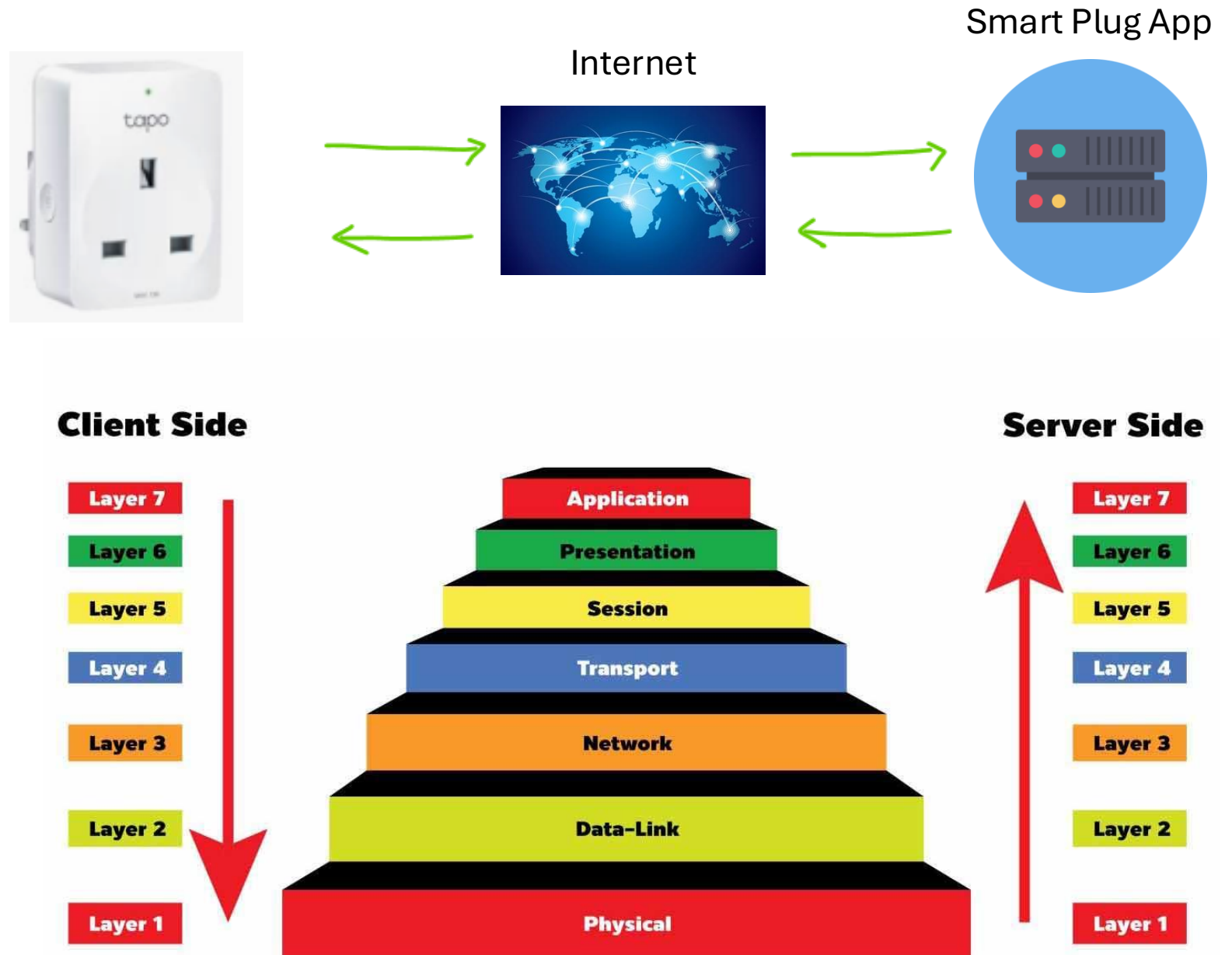
Attacks:

- Man-in-the-Middle
- unauthorized login
- DOS
- Etc.

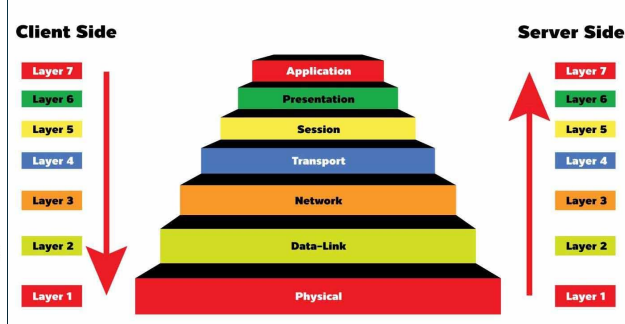
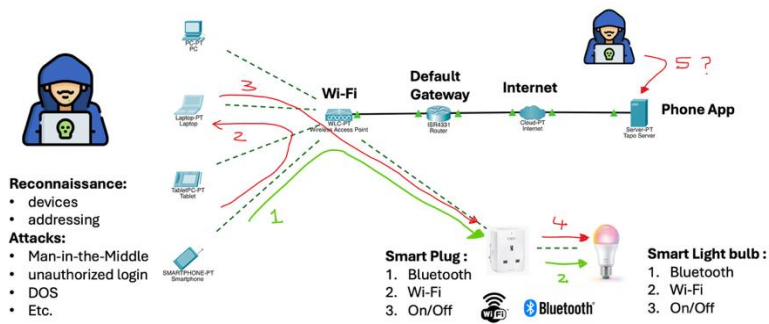


OSI model

describes seven layers that computer systems use to communicate over a network

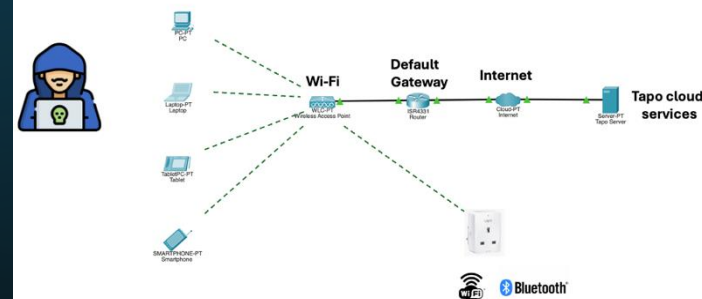


Addressing



OSI Layer	Protocol	Detail	Example
Layer 5-7 (Application)	Data	Application	e.g. Smart plug app
Layer 4 (Transport)	TCP or UDP	Port number	e.g. 443, 8009 etc
Layer 3 (Network)	IPv4 or IPv6	IP addressing	e.g. 192.168.1.10 (32 bits) e.g. 2001:0000:130F:0000:0000:09C0:876A:130B (128 bits)
Layer 2 (Data Link)	Wi-Fi	MAC addressing	e.g. 00-B0-D0-63-C2-26 (48 bits, 12 Hexadecimal)
	Wi-Fi	SSID	<i>IPL Wifi</i>
	Bluetooth	MAC addressing	e.g. 00:11:22:33:FF:EE (48 bits, 12 Hexadecimal)
Layer 1 (Physical)	Wi-Fi & Bluetooth	Radio Frequency	2.4 and 5 GHz bands

The Hacking challenges...



OSI Layer	Protocol	Find Smart plug, bulb, app...	Hack Success?
Layer 5-7	Application	Authentication?	✓ or ✗
Layer 4	TCP or UDP	Port number?	✓ or ✗
Layer 3	IPv4 or IPv6	IP address?	✓ or ✗
Layer 2	Wi-Fi / Bluetooth	MAC address?	✓ or ✗
	Wi-Fi / Bluetooth	Authentication?	IPL wi-fi ✓
	Wi-Fi / Bluetooth	SSID?	IPL wi-fi ✓
Layer 1	Wi-Fi / Bluetooth	Frequency?	IPL wi-fi ✓

Enjoy!

